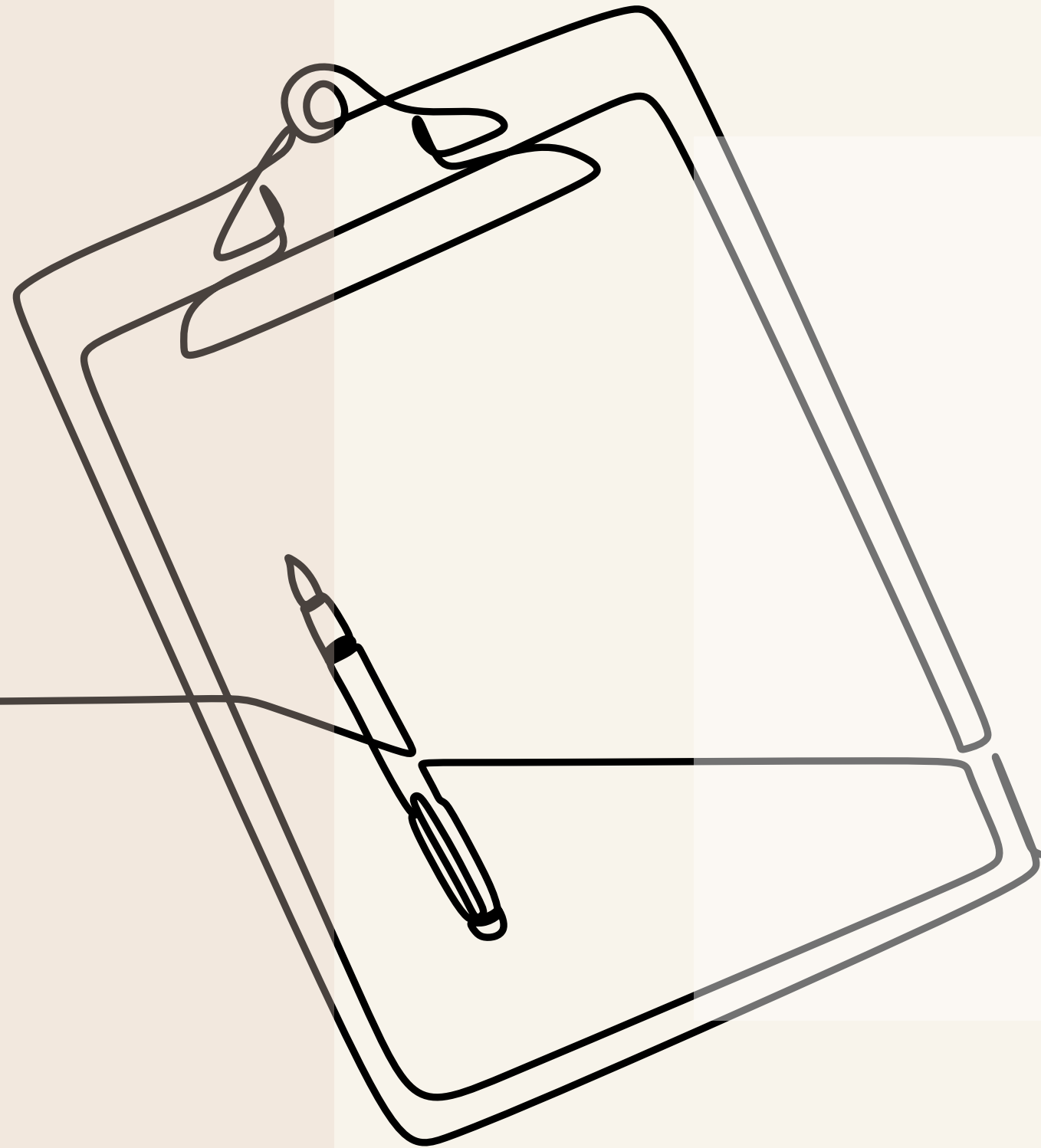




WATERFALL MODEL

PRESENTED BY:

JUSTIN ATCHERA

SARINO ANSTY

KENN SONIO

METHODOLOGY CHOSEN



Methodology: Waterfall Model

- A step-by-step approach where each phase is completed in order before moving to the next.

Waterfall Model



The Waterfall Model is a step-by-step software development approach where each phase is completed and approved before moving to the next.

Why this model exists

It exists to provide a clear and organized development process, ensure proper documentation, and make project progress easier to manage when requirements are clear and stable.

HOW WATERFALL WORKS

The project begins with requirements, followed by design, development, testing, deployment, and maintenance. Each phase must be completed and approved before moving to the next, which makes the process organized and easy to track.





UNIQUE FEATURES VS OTHER SDLC MODELS

Unique Features of Waterfall Model

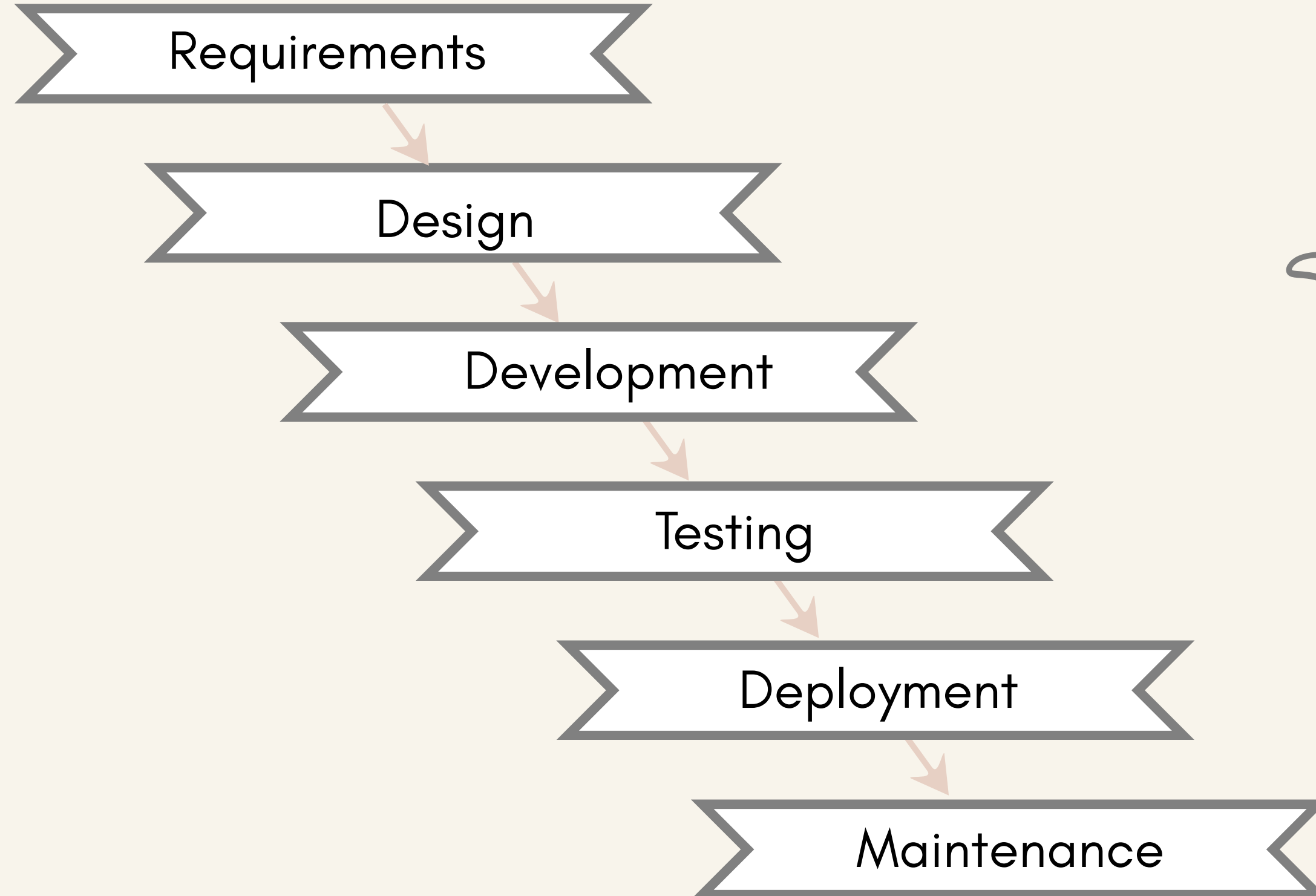
- Strict sequence of phases
- Strong documentation and approvals
- Clear milestones and deliverables
- Easy progress tracking



Compared to Agile:

- Waterfall is linear and phase-based
- Agile is iterative and sprint-based

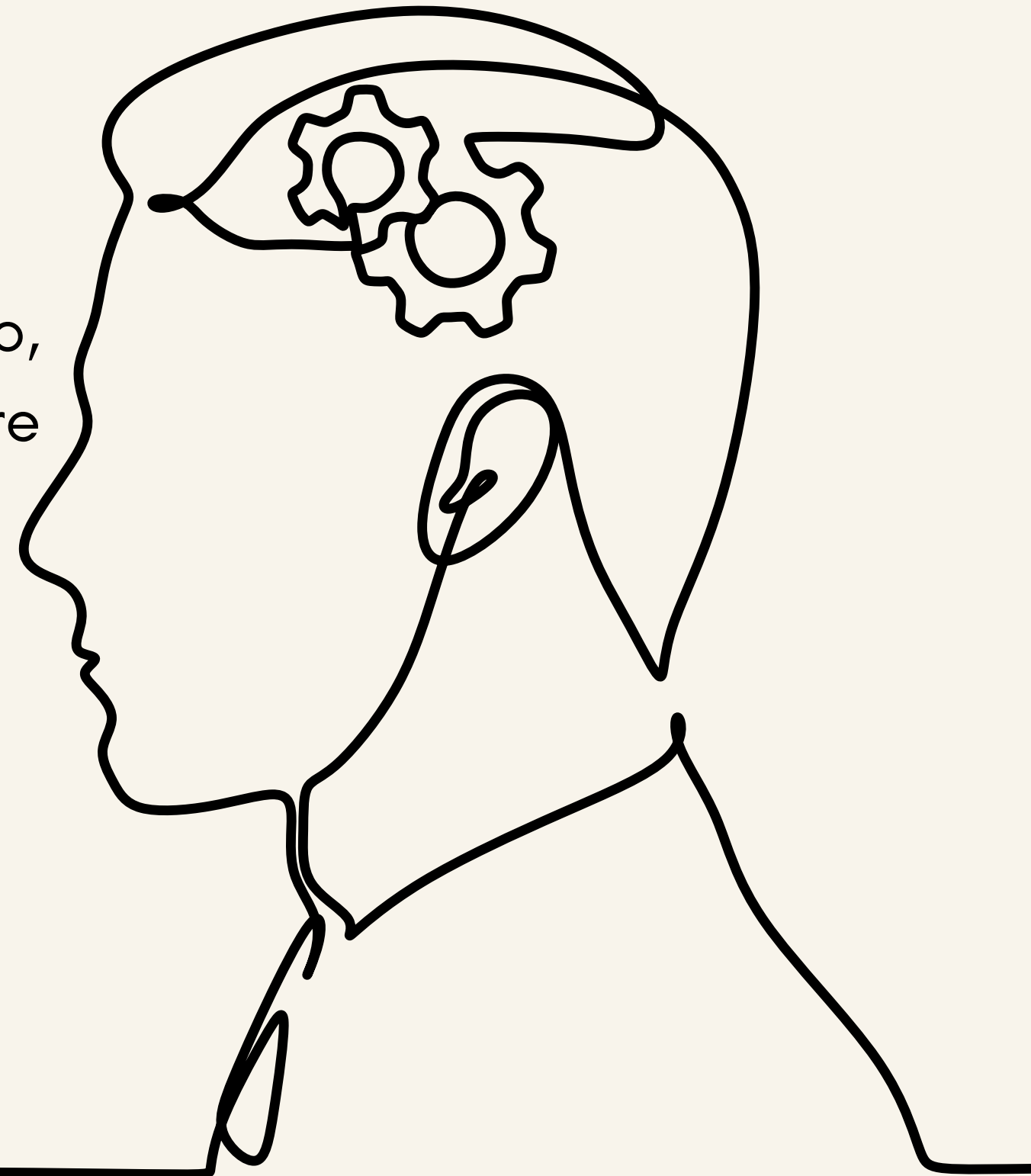
VISUAL DIAGRAM



PHASE 1: REQUIREMENTS

This phase identifies and defines what the system must do, including its features, user needs, and requirements before design and development begin.

- FRD / NFRD
- User Stories
- RTM (Requirement mapping)
- Waterfall rule: Requirements must be approved before design



PHASE 2: SYSTEM DESIGN

This phase creates the blueprint of the system, including the user interface, system architecture, and database/data model design.

- User Flow + Wireframes/Prototype (Figma)
- UI Style Guide
- Architecture Diagram (App + Firebase)
- Firestore Data Model + Security Rules plan



PHASE 3: DEVELOPMENT (CODING)

This phase builds the actual application by coding the planned modules and configuring the required backend services.

- React Native Android app modules (auth, tracking, meals, macros, dashboard)
- Firebase setup (Auth, Firestore, FCM, Security Rules)
- Waterfall rule: Coding starts after final design



PHASE 4: TESTING

This phase verifies that the system works correctly, meets the requirements, and is ready for deployment.

- Test Plan
- Unit / Integration / System Testing
- UAT Sign-off
- Defect Closure Report



PHASE 5: DEPLOYMENT

This phase prepares and releases the final build of the application for actual use or demonstration.

- Signed APK/AAB release build
- Smoke test results
- Rollback/Hotfix plan



PHASE 6: MAINTENANCE

This phase focuses on fixing issues, improving the system, and applying updates after release.

- Bug fixes, updates, improvements
- Documentation updates if needed



How Waterfall Handles the Phases

Waterfall handles the phases in a step-by-step sequence, where one phase is finished first before moving to the next. This makes the process clear and structured, but less flexible to changes.



Why Teams Choose Waterfall

- Clear scope and predictable workflow
- Easy tracking of deliverables
- Good for documentation-heavy projects (capstone)
- Works well when requirements are stable



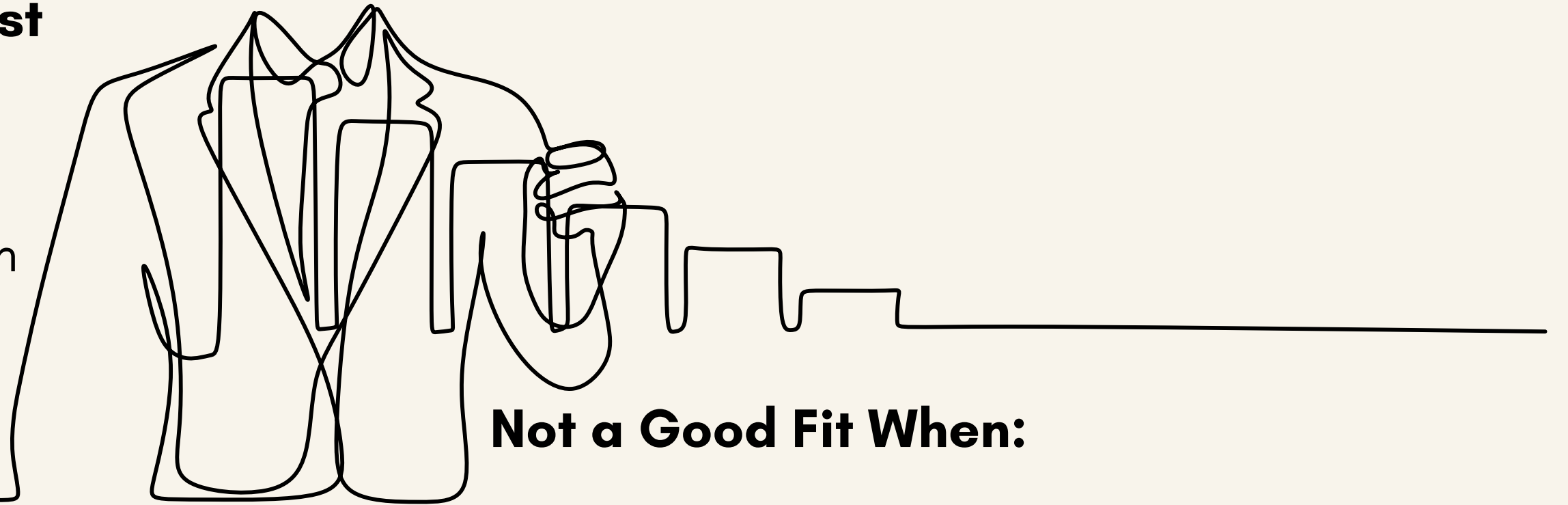
Limitations or Challenges of Waterfall

- Changes are difficult once a phase is completed
- Working software is seen later in the project
- Errors in early phases may affect later phases
- Rework can be costly if requirements are wrong at the start



Situations Where Waterfall Works Best

- Stable requirements
- Projects needing formal documentation
- Fixed timeline and scope (school projects)



Not a Good Fit When:

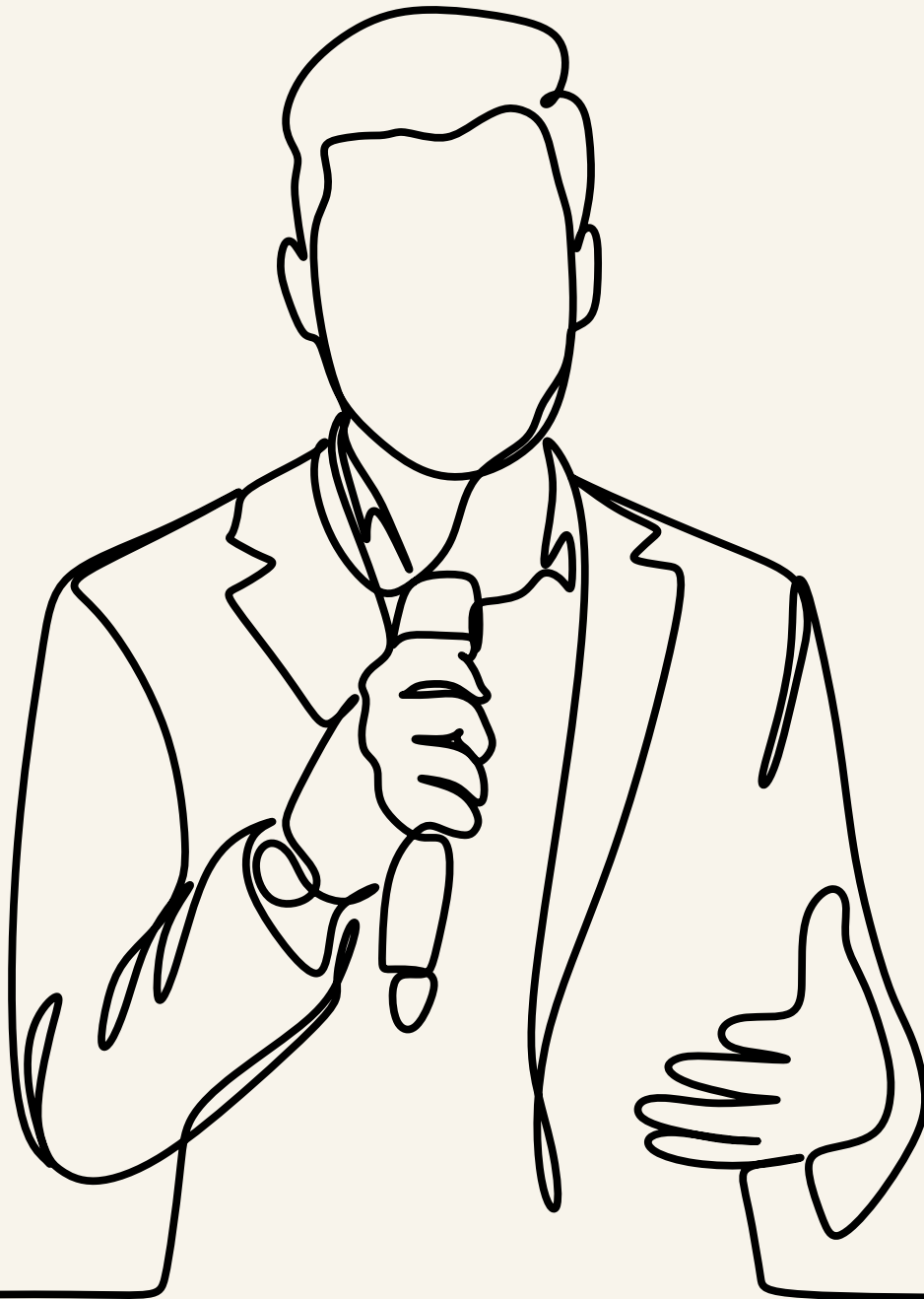
- Requirements change frequently
- Frequent releases/weekly feedback needed
- High uncertainty projects

SIMPLE CASE STUDY / SCENARIO

Scenario: A small fitness coaching team wants an Android mobile application where users can log workouts, plan meals, and monitor daily progress in one place. The team also wants reminders and secure data storage using Firebase.

Project Situation:

- The core features are already identified, such as login, profile/goals, fitness logging, meal planning, macro tracking, and dashboard.
- The project follows a fixed academic timeline, so the team needs a clear phase-by-phase workflow.
- The project requires complete documentation and approved outputs for capstone submission.



How the Waterfall Model Guides Our Project

- **Requirements:** finalize app features, goals, and documents
- **Design:** create Figma prototype, architecture, and Firestore data model
- **Development:** build React Native screens and configure Firebase services
- **Testing:** perform unit, integration, and end-to-end testing
- **Deployment:** generate APK/AAB and perform smoke testing
- **Maintenance:** fix issues and improve the app after feedback





Difference from Agile

Waterfall follows a linear and sequential flow, where each phase must be completed before moving to the next. Agile, on the other hand, is iterative and sprint-based, allowing frequent changes and continuous feedback throughout development.

The background is a light cream color. In the top-left corner, there are thin, light pink wavy lines. In the top-right corner, there are thick, dark grey wavy lines. In the bottom-left corner, there are thick, dark grey wavy lines. In the bottom-right corner, there are thin, light pink wavy lines.

Thank You